

Noncritical Type 0B String Theory and Its Matrix-Model Dual

Setup, conventions, and first benchmarks

StringMC project

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Abstract

This note specifies a new target for the StringMC project: the duality between two-dimensional noncritical type 0B string theory and double-scaled Hermitian matrix quantum mechanics with a symmetrically filled two-sided Fermi sea. We adopt explicitly the conventions of Balthazar–Rodriguez–Yin (BRY), including $\alpha' = 2$, their delta-normalized physical states, and their exact matrix-model energy and coupling dictionary. We identify the genus-one logarithm as the first vacuum normalization benchmark and separate carefully the sphere data supplied by BRY from the additional supermoduli measure needed at genus one and genus two.

Contents

1	Target and baseline	2
2	Worldsheet description	2
2.1	$\widehat{c} = 1$ matter and super-Liouville theory	2
2.2	Type 0B projection and spectrum	3
2.3	Physical vertex operators, pictures, and sphere three-point data	4
3	Sphere four-point superconformal blocks	7
3.1	Labels, weights, and local series	7
3.2	The c -recursion	8
3.3	Elliptic representation and code interface	9
3.4	Nonchiral correlators and the BRY four-tachyon density	10
3.5	First regulated $1_R \rightarrow 3_R$ benchmark	11
4	Matrix quantum mechanics	13
4.1	Double-scaled Hamiltonian	13
4.2	Two-sided basis and exact dictionary	13
5	Duality dictionary	13
6	Genus-one path integral in BRY variables	13
6.1	Even spin structures and the modular orbit integral	14
6.2	Odd spin structure and direct odd-supermodulus integration	15
6.3	The 0B and fluxless 0A spin sums	15

6.4	Liouville wall and the regulator ledger	16
6.5	Douglas convention and the matrix variable	17
7	Benchmark ladder	17
8	Worksheet literature for the genus-one and genus-two program	18
8.1	Genus-one measure source	18
8.2	Genus two: conformal data and measure	19
8.3	Vertical integration at genus two	19
8.4	Result of the literature audit	20
9	Deferred choices and safeguards	20
10	Immediate next step	20

1 Target and baseline

The target is the proposed exact duality

$$\boxed{\text{2D noncritical type 0B string theory} \longleftrightarrow \text{two-sided double-scaled MQM}} \quad (1)$$

developed in Refs. [1, 2]. Our master convention is Ref. [3]. The essential distinction from the bosonic $c = 1$ matrix model is the vacuum: both sides of an even potential are filled to the same Fermi level. The resulting state has a nonperturbatively stable completion and supports two collective modes, which map to the two propagating type 0B fields.

The initial comparison is defined by

$$\begin{aligned} b = 1, \quad Q = b + b^{-1} = 2, \quad \ell_B \equiv \sqrt{\alpha'/2}, \quad [\ell_B]_{\text{BRY}} = 1, \\ q_{\text{R-R}} = \tilde{q}_{\text{R-R}} = 0, \quad E_F = -\mu_F, \quad \mu_F > 0. \end{aligned} \quad (2)$$

with equal regulated fillings $N_L = N_R$. For the finite-radius benchmark, Euclidean time is compactified as

$$X \sim X + 2\pi R_{\text{phys}}. \quad (3)$$

The BRY notation $\alpha' = 2$ is a choice of units: $\alpha' = 2\ell_B^2$ followed by $[\ell_B]_{\text{BRY}} = 1$. It is not an equality between a dimensionful quantity and the pure number two. Every formula imported from another source is translated to this $\alpha' = 2$ convention before use.

2 Worksheet description

2.1 $\hat{c} = 1$ matter and super-Liouville theory

The matter multiplet is the Lorentzian free field $(X^0, \psi^0, \tilde{\psi}^0)$. In the BRY component normalization, the $N = 1$ super-Liouville action is

$$\begin{aligned} S_{\text{SL}} = \int d^2z \left[\frac{1}{2\pi} \left(\partial\phi \bar{\partial}\phi + \psi \bar{\partial}\psi + \tilde{\psi} \partial\tilde{\psi} \right) \right. \\ \left. + 2i\mu_L b^2 \psi \tilde{\psi} e^{b\phi} + 2\pi b^2 \mu_L^2 e^{2b\phi} \right]. \end{aligned} \quad (4)$$

The linear-dilaton slope and ordinary central charge are

$$Q = b + b^{-1}, \quad c_{\text{SL}} = \frac{3}{2}(1 + 2Q^2). \quad (5)$$

At $b = 1$, $Q = 2$ and $c_{\text{SL}} = 27/2$; together with the free timelike multiplet this gives an $N = 1$, $c = 15$ matter SCFT.

We reserve ϕ for the Liouville boson and $\varphi, \tilde{\varphi}$ for the bosonized superghosts. Throughout, h and $\tilde{h}$ denote the holomorphic and antiholomorphic conformal weights. In the $\alpha' = 2$ convention the elementary holomorphic weights are

$$h[e^{\pm i\omega X^0}] = -\frac{\omega^2}{2}, \quad h[e^{q\varphi}] = -\frac{1}{2}q(q+2), \quad h[c] = -1, \quad h[\sigma^0] = h[\mu^0] = \frac{1}{16}, \quad (6)$$

with identical formulas in the antiholomorphic sector. Thus $h[e^{-\varphi}] = 1/2$ and $h[e^{-\varphi/2}] = 3/8$. Equation (6) fixes in one place the timelike, superghost, and spin-field conventions used below.

The super-Liouville wall cuts off the strong-coupling end of the linear-dilaton direction. The weak-coupling asymptotic end supplies the scattering region. BRY print the glyph μ both for the Liouville interaction in Eq. (4) and for the matrix-model Fermi depth. We denote them by μ_{L} and μ_{F} respectively. They are not identified inside the genus-one derivation; the comparison map $\mu_{\text{L}} = \kappa\mu_{\text{F}}^p$ is introduced only afterward.

2.2 Type 0B projection and spectrum

The nonchiral type 0B GSO projection retains

$$(\text{NS}-, \text{NS}-) \oplus (\text{NS}+, \text{NS}+) \oplus (\text{R}+, \text{R}+) \oplus (\text{R}-, \text{R}-). \quad (7)$$

There are no spacetime fermions. The propagating fields are an NS–NS scalar T and an R–R axion A . The field traditionally called the “tachyon” is massless in the two-dimensional linear-dilaton background. Discrete states also occur at special momenta, but they are not part of the first vacuum-amplitude benchmark.

For $P \geq 0$, BRY normalize the NS and R super-Liouville primaries by

$$\begin{aligned} h_{\text{NS}}(P) &= \frac{1}{2} \left(\frac{Q^2}{4} + P^2 \right) = \frac{1 + P^2}{2}, \\ h_{\text{R}}(P) &= \frac{1}{16} + \frac{1}{2} \left(\frac{Q^2}{4} + P^2 \right) = \frac{1}{16} + \frac{1 + P^2}{2}, \end{aligned} \quad (8)$$

Here h_{R} is the weight of the *Liouville* Ramond operator; the timelike spin or disorder field contributes a further $1/16$. Hence the combined matter Ramond operator used in the physical axion vertex has

$$h[\mathcal{A}_P^\pm] = \tilde{h}[\mathcal{A}_P^\pm] = \frac{1}{16} + h_{\text{R}}(P) = \frac{1}{8} + \frac{1}{2}(1 + P^2). \quad (9)$$

The reflection phases are fixed by the weak-coupling, $\phi \rightarrow -\infty$, asymptotics

$$V_P \sim S_{\text{NS}}(P)^{-1/2} e^{(Q/2+iP)\phi} + S_{\text{NS}}(P)^{1/2} e^{(Q/2-iP)\phi}, \quad (10)$$

$$V_P^{\text{R},+} \sim \sigma^1 \left[S_{\text{R}}(P)^{-1/2} e^{(Q/2+iP)\phi} + S_{\text{R}}(P)^{1/2} e^{(Q/2-iP)\phi} \right], \quad (11)$$

$$V_P^{\text{R},-} \sim \mu^1 \left[S_{\text{R}}(P)^{-1/2} e^{(Q/2+iP)\phi} - S_{\text{R}}(P)^{1/2} e^{(Q/2-iP)\phi} \right]. \quad (12)$$

Here σ^1 and μ^1 are the spin and disorder fields of the asymptotic Liouville Majorana fermion. The relative minus sign in Eq. (12) is part of the Ramond phase convention, not an in/out label. The explicit functions S_{NS} and S_{R} are not needed in the BRY sphere amplitudes.

With those phases, the delta-function normalization is

$$\langle V_{P_1}(z, \bar{z}) V_{P_2}(0) \rangle = \frac{\pi \delta(P_1 - P_2)}{|z|^{2(h_1 + h_2)}}. \quad (13)$$

For either Ramond family $\epsilon = \pm$, the corresponding normalization is

$$\langle V_{P_1}^{\text{R}, \epsilon}(z, \bar{z}) V_{P_2}^{\text{R}, \epsilon}(0) \rangle = \frac{\pi \delta(P_1 - P_2)}{|z|^{2(h_1^{\text{R}} + h_2^{\text{R}})}}. \quad (14)$$

In the modular-invariant bosonic-CFT description, $V_P^{\text{R}, +}$ is a local lowest-weight operator, whereas $V_P^{\text{R}, -}$ is a defect operator attached to the $\mathbb{Z}_2$ fermion-parity topological line. Thus the two symbols do not denote two ordinary local-primary families. The equal-family Ramond correlators below contain either zero or two defect endpoints; a mixed $+/-$ correlator is not an ordinary local sphere correlator in the BRY data. Equations (13) and (14), together with the standard free-field, spin-field, and ghost inner products, are what is meant below by the delta-normalized asymptotic string states. They should not be confused with a two-punctured sphere vacuum diagram, which retains a conformal-Killing volume. For the timelike zero mode we use the Fourier convention

$$\int \frac{dX_0^0}{2\pi} e^{i(\omega - \omega_1 - \omega_2)X_0^0} = \delta(\omega - \omega_1 - \omega_2). \quad (15)$$

Accordingly, the energy delta function below carries no additional 2π ; that factor is absorbed into the zero-mode path-integral measure.

2.3 Physical vertex operators, pictures, and sphere three-point data

The incoming and outgoing physical operators are

$$T_\omega^\pm = g_s c \tilde{c} e^{-\varphi - \tilde{\varphi}} e^{\pm i\omega X^0} V_{P=\omega}, \quad (16)$$

$$A_\omega^\pm = g_s \omega c \tilde{c} e^{-\varphi/2 - \tilde{\varphi}/2} e^{\pm i\omega X^0} \mathcal{A}_{P=\omega}^\pm, \quad (17)$$

$$\mathcal{A}_P^\pm = \frac{1}{\sqrt{2}} \left(\sigma^0 V_P^{\text{R}, +} \pm \mu^0 V_P^{\text{R}, -} \right). \quad (18)$$

Equivalently, their weak-coupling wave functions are

$$\begin{aligned} \mathcal{A}_P^+ &\sim \text{S}^+ \tilde{\text{S}}^+ S_{\text{R}}(P)^{-1/2} e^{(Q/2 + iP)\phi} + \text{S}^- \tilde{\text{S}}^- S_{\text{R}}(P)^{1/2} e^{(Q/2 - iP)\phi}, \\ \mathcal{A}_P^- &\sim \text{S}^- \tilde{\text{S}}^- S_{\text{R}}(P)^{-1/2} e^{(Q/2 + iP)\phi} + \text{S}^+ \tilde{\text{S}}^+ S_{\text{R}}(P)^{1/2} e^{(Q/2 - iP)\phi}, \end{aligned} \quad (19)$$

where $\text{S}^\pm$ and $\tilde{\text{S}}^\pm$ are the spin fields of the combined timelike–Liouville fermion system, normalized so that

$$\text{S}^\pm \tilde{\text{S}}^\pm = \frac{1}{\sqrt{2}} \left(\sigma^0 \sigma^1 \pm \mu^0 \mu^1 \right). \quad (20)$$

Here $T_\omega^\pm$ is an NS–NS vertex in picture $(-1, -1)$ and $A_\omega^\pm$ is an R–R vertex in picture $(-\frac{1}{2}, -\frac{1}{2})$. For $\omega > 0$, the superscript has the fixed scattering interpretation

$$+ = \text{incoming}, \quad - = \text{outgoing}. \quad (21)$$

Thus an incoming state carries $e^{+i\omega X^0}$ and an outgoing state carries $e^{-i\omega X^0}$ in these conventions. The symbols $\pm$ on the Liouville operators $V_P^{\text{R},\pm}$ instead label the two Ramond families in Eq. (7); they are not in/out labels. The combination in Eq. (18) correlates these two distinct uses of a sign. BRY choose their phases so that Eqs. (13) and (14) hold and all structure constants below are real for $P_i \geq 0$. No additional leg-pole factor is included.

The BRST mass-shell condition is $P = \omega$. Before ghost dressing, the left-moving matter weights are

$$\begin{aligned} h_T^{\text{matter}}(P, \omega) &= -\frac{\omega^2}{2} + h_{\text{NS}}(P) = \frac{1}{2} + \frac{P^2 - \omega^2}{2}, \\ h_A^{\text{matter}}(P, \omega) &= -\frac{\omega^2}{2} + h[\mathcal{A}_P^\pm] = \frac{5}{8} + \frac{P^2 - \omega^2}{2}. \end{aligned} \quad (22)$$

At $P = \omega$, adding $e^{-\varphi}$ to the NS operator or $e^{-\varphi/2}$ to the R operator gives weight one; multiplication by c then gives weight zero. The same check holds on the right, so the unintegrated physical vertices in Eqs. (16) and (17) have total weight $(0, 0)$.

To state the three-point data independently of the ghost dressing, define the level-1/2 descendants and the top NS component by

$$\Lambda_P = G_{-1/2}V_P, \quad \tilde{\Lambda}_P = \tilde{G}_{-1/2}V_P, \quad W_P = G_{-1/2}\tilde{G}_{-1/2}V_P. \quad (23)$$

BRY normalize the holomorphic supercurrent modes by

$$\{G_r, G_s\} = 2L_{r+s} + \frac{c}{12}(4r^2 - 1)\delta_{r,-s}, \quad (24)$$

and analogously in the antiholomorphic sector. The top component has two-point function

$$\langle W_{P_1}(z, \bar{z})W_{P_2}(0) \rangle = \frac{\pi(2h_1)^2\delta(P_1 - P_2)}{|z|^{2(h_1+h_2+1)}}. \quad (25)$$

Writing $z_{ij} = z_i - z_j$, the two independent NS three-point functions are

$$\langle V_{P_1}(z_1)V_{P_2}(z_2)V_{P_3}(z_3) \rangle = \frac{C(P_1, P_2, P_3)}{|z_{12}|^{2(h_1+h_2-h_3)}|z_{13}|^{2(h_1+h_3-h_2)}|z_{23}|^{2(h_2+h_3-h_1)}}, \quad (26)$$

$$\langle W_{P_1}(z_1)V_{P_2}(z_2)V_{P_3}(z_3) \rangle = \frac{\tilde{C}(P_1, P_2, P_3)}{|z_{12}|^{2(h_1+h_2-h_3+1/2)}|z_{13}|^{2(h_1+h_3-h_2+1/2)}|z_{23}|^{2(h_2+h_3-h_1-1/2)}}. \quad (27)$$

For two Ramond fields and one NS field, let h_i^{R} denote the Ramond weights. The nonzero equal-family correlators are

$$\left\langle V_{P_3}(z_3)V_{P_2}^{\text{R},\epsilon}(z_2)V_{P_1}^{\text{R},\epsilon}(z_1) \right\rangle = \frac{C_\epsilon(P_1, P_2; P_3)}{|z_{12}|^{2(h_1^{\text{R}}+h_2^{\text{R}}-h_3)}|z_{13}|^{2(h_1^{\text{R}}+h_3-h_2^{\text{R}})}|z_{23}|^{2(h_2^{\text{R}}+h_3-h_1^{\text{R}})}}, \quad \epsilon = \pm, \quad (28)$$

where

$$C_\pm = \frac{1}{2}(C_{\text{even}} \pm C_{\text{odd}}). \quad (29)$$

The coefficients are completely explicit at $b = 1$. Define

$$\begin{aligned} \Upsilon_{\text{NS}}(x) &= \frac{\Gamma(x/2)}{\Gamma(1-x/2)}\Upsilon_1(x/2)^2, & \Upsilon_{\text{R}}(x) &= \Upsilon_1\left(\frac{x+1}{2}\right)^2, \\ N_{\text{NS}}(P) &= \frac{\Gamma(1+iP)}{\Gamma(1-iP)}\Upsilon_{\text{NS}}(2iP), & N_{\text{R}}(P) &= \frac{\Gamma(\frac{1}{2}+iP)}{\Gamma(\frac{1}{2}-iP)}\Upsilon_{\text{R}}(2iP), \end{aligned} \quad (30)$$

The $b = 1$ Barnes-double-gamma convention is fixed, including its overall normalization, by [4]

$$\begin{aligned} G(1) &= 1, & G(x+1) &= \Gamma(x)G(x), \\ \Gamma_1(x) &= \frac{(2\pi)^{(x-1)/2}}{G(x)}, & \Gamma_1(x+1) &= \frac{\sqrt{2\pi}}{\Gamma(x)}\Gamma_1(x), \\ \Upsilon_1(x) &= \frac{1}{\Gamma_1(x)\Gamma_1(2-x)}, & \Upsilon_1(1) &= 1. \end{aligned} \quad (31)$$

Here G is the Barnes G -function on its standard analytic continuation. Equivalently, Γ_1 is meromorphic with poles at the nonpositive integers, while Υ_1 is entire, has zeros at $x \in \mathbb{Z} \setminus \{1\}$, and obeys

$$\Upsilon_1(2-x) = \Upsilon_1(x), \quad \Upsilon_1(x+1) = \frac{\Gamma(x)}{\Gamma(1-x)}\Upsilon_1(x). \quad (32)$$

The power of 2π in Eq. (31) uses the real $\log(2\pi)$, so it introduces no additional complex-log branch choice. At $P = 0$, and whenever individual Gamma factors appear singular, the composite expressions in Eqs. (30)–(37) are defined by analytic continuation and their continuous $P \rightarrow 0^+$ limit, not by separately evaluating the singular factors.

With

$$P_\Sigma = P_1 + P_2 + P_3, \quad \Delta_i = P_j + P_k - P_i, \quad \{i, j, k\} = \{1, 2, 3\}, \quad (33)$$

the two NS coefficients are

$$C(P_1, P_2, P_3) = \frac{i}{2} \frac{\prod_{j=1}^3 N_{\text{NS}}(P_j)}{\Upsilon_{\text{NS}}(1+iP_\Sigma) \prod_{i=1}^3 \Upsilon_{\text{NS}}(1+i\Delta_i)}, \quad (34)$$

$$\tilde{C}(P_1, P_2, P_3) = i \frac{\prod_{j=1}^3 N_{\text{NS}}(P_j)}{\Upsilon_{\text{R}}(1+iP_\Sigma) \prod_{i=1}^3 \Upsilon_{\text{R}}(1+i\Delta_i)}. \quad (35)$$

For P_1, P_2 Ramond and P_3 NS, set $\mathcal{N} = N_{\text{R}}(P_1)N_{\text{R}}(P_2)N_{\text{NS}}(P_3)$. Then

$$C_{\text{even}}(P_1, P_2; P_3) = -\frac{i}{2} \frac{\mathcal{N}}{\Upsilon_{\text{R}}(1+iP_\Sigma)\Upsilon_{\text{R}}(1+i\Delta_3)\Upsilon_{\text{NS}}(1+i\Delta_1)\Upsilon_{\text{NS}}(1+i\Delta_2)}, \quad (36)$$

$$C_{\text{odd}}(P_1, P_2; P_3) = -\frac{i}{2} \frac{\mathcal{N}}{\Upsilon_{\text{NS}}(1+iP_\Sigma)\Upsilon_{\text{NS}}(1+i\Delta_3)\Upsilon_{\text{R}}(1+i\Delta_1)\Upsilon_{\text{R}}(1+i\Delta_2)}. \quad (37)$$

These coefficients are real for $P_i \geq 0$ in the BRY phase convention. Under analytic continuation that reflects one momentum, C and $\tilde{C}$ change sign; reflecting the NS momentum changes the sign of both C_{even} and C_{odd} , while reflecting either Ramond momentum exchanges their even and odd labels. All descendant correlators are then fixed by superconformal Ward identities.

The holomorphic PCO used in their sphere calculation is

$$\chi = -\frac{1}{2}e^\varphi \left(-i\psi^0 \partial X^0 + G^{\text{L}} \right) + \text{ghost terms}, \quad (38)$$

and similarly in the antiholomorphic sector. At higher genus this local operator represents odd-modulus integration only after a global supermoduli or vertical-integration prescription has been supplied. For the tree-level NS amplitudes, BRY place a holomorphic and an antiholomorphic PCO on a tachyon insertion. Suppressing the spectator $c\tilde{c}$ factor, the resulting picture-raised operator is

$$\begin{aligned} \tilde{\chi}\chi T_\omega^\pm &= \frac{ig_s}{4} [\omega^2 \psi^0 \tilde{\psi}^0 e^{\pm i\omega X^0} V_\omega + e^{\pm i\omega X^0} W_\omega \\ &\quad \pm \omega \tilde{\psi}^0 e^{\pm i\omega X^0} \Lambda_\omega \mp \omega \psi^0 e^{\pm i\omega X^0} \tilde{\Lambda}_\omega] + \text{ghost terms}. \end{aligned} \quad (39)$$

The explicit i in Eq. (39) is BRY's cocycle phase: combining the separately bosonized left- and right-moving factors $e^{-\varphi}e^{-\tilde{\varphi}}$ into $e^{-\varphi-\tilde{\varphi}}$ produces this extra factor. It is already included in the amplitude convention below. Their ordered integration over a complex-conjugate pair of Grassmann-odd moduli inserts $-i\tilde{\chi}\chi$. This sign is included in the sphere amplitude below.

For completeness, distinguish the CFT three-point functions above from the on-shell three-string amplitude. The total picture on the sphere is $(-2, -2)$. For one incoming and two outgoing right-side modes $\mathcal{R} = T + A$, place the incoming vertex at infinity and the outgoing vertices at 1 and 0. In the BRY cocycle and odd-modulus convention, with C_{S^2} denoting the sphere-topology normalization,

$$S_{1\mathcal{R}\rightarrow 2\mathcal{R}}(\omega, \omega_1, \omega_2) = i\delta(\omega - \omega_1 - \omega_2)g_s^3C_{S^2}\left[\frac{1}{4}\tilde{C}(\omega, \omega_1, \omega_2) + \frac{1}{2}\omega_1\omega_2C_{\text{even}}(\omega_1, \omega_2; \omega) + \frac{1}{2}\omega\omega_2C_{\text{odd}}(\omega, \omega_2; \omega_1) + \frac{1}{2}\omega\omega_1C_{\text{odd}}(\omega, \omega_1; \omega_2)\right]. \quad (40)$$

On the energy-conserving locus $\omega = \omega_1 + \omega_2$, the bracket collapses to $\omega\omega_1\omega_2$, so after removing the delta function

$$\boxed{\mathcal{A}_{1\mathcal{R}\rightarrow 2\mathcal{R}}^{S^2} = ig_s^3C_{S^2}\omega\omega_1\omega_2 = \frac{4i}{\mu_F}\omega\omega_1\omega_2.} \quad (41)$$

The same formula holds for three left-side modes $\mathcal{L} = T - A$. Every sphere three-point amplitude containing both an $\mathcal{L}$ and an $\mathcal{R}$ mode vanishes. In particular, in the T, A basis the zero-flux selection rule eliminates amplitudes with an odd number of axions.

The odd spin structure contributes to the type 0B torus amplitude. It is therefore essential to retain the complete spin-structure sum: using only the three even structures gives the wrong coefficient for the two massless spacetime fields.

The three-punctured-sphere ledger above and the four-punctured-sphere block layer below are intentionally separate. The former fixes external-state and pants normalizations; the latter computes chiral NS superconformal blocks but does not yet perform the modulus integral of a four-string amplitude.

3 Sphere four-point superconformal blocks

This section fixes the code convention for the $N = 1$ super-Virasoro block on a four-punctured sphere, following BRY and the c -recursion of Refs. [5, 6, 7, 3]. Only NS external and exchanged representations are included at this stage. The pointwise NS structure-constant contraction and the integral over real internal Liouville momentum are included. Ramond blocks, the integral over the sphere modulus and its counterterms, analytic continuation to physical scattering kinematics, and every higher-genus sewing problem remain outside the scope of the present implementation.

3.1 Labels, weights, and local series

Parametrize the super-Liouville central charge and weights by

$$c = \frac{3}{2}(1 + 2Q^2), \quad Q = b + b^{-1}, \quad h_i = \frac{1}{2}\left(\frac{Q^2}{4} - a_i^2\right), \quad h = \frac{1}{2}\left(\frac{Q^2}{4} + P^2\right). \quad (42)$$

For the type 0B theory, $b = 1$, $Q = 2$, $c = 27/2$, and an external super-Liouville primary of momentum P_i has $h_i = (1 + P_i^2)/2$. At punctures 2 and 3 the external state may be either a

primary or its level-1/2 descendant. Write

$$\star^0 h_i \equiv h_i, \quad \star^1 h_i \equiv *h_i, \quad \underline{h}_i \equiv h_i + \frac{\delta_i}{2}, \quad \delta_i \in \{0, 1\}, \quad i = 2, 3. \quad (43)$$

Thus there are eight chiral blocks $\mathcal{F}^{e/o}(h_4, \star^{\delta_3} h_3, \star^{\delta_2} h_2, h_1; h|z)$, where e/o selects integer/half-integer levels in the internal NS module. The cross ratio and our local series are

$$\begin{aligned} z &= \frac{z_{12}z_{34}}{z_{13}z_{24}}, \\ \mathcal{F}^e(z) &= z^{h-h_1-\underline{h}_2} \left[1 + \sum_{m=1}^{\infty} z^m \mathfrak{F}_m \right], \\ \mathcal{F}^o(z) &= z^{h-h_1-\underline{h}_2} \sum_{n=0}^{\infty} z^{n+1/2} \mathfrak{F}_{n+1/2}. \end{aligned} \quad (44)$$

The fraktur coefficients specify the chiral phase used in the code: they are the signed coefficients produced by BRY's ancillary notebook, including the minus signs in the half-integer seeds below, with no second overall sign inserted in the last line of Eq. (44). BRY's displayed block convention instead places an extra minus sign in front of the odd series when $(\delta_3, \delta_2) = (1, 1)$. This is a holomorphic phase choice, which the scalar decomposition into $|\mathcal{F}|^2$ does not fix; the code choice is made explicit because it reproduces BRY's executable crossing benchmark.

3.2 The c -recursion

For $r \geq 2$, $s \geq 1$, and $r + s$ even, define

$$\begin{aligned} b_{r,s}(h)^2 &= -\frac{4h + rs - 1 + \sqrt{16h^2 + 8(rs - 1)h + (r - s)^2}}{r^2 - 1}, \\ c_{r,s}(h) &= \frac{15}{2} + 3b_{r,s}(h)^2 + 3b_{r,s}(h)^{-2}, \\ A_{r,s}(h) &= \frac{1}{2} \prod_{\substack{p=1-r, \dots, r \\ q=1-s, \dots, s \\ (p,q) \neq (0,0), (r,s) \\ p+q \in 2\mathbb{Z}}} \frac{\sqrt{2}}{pb_{r,s}(h) + qb_{r,s}(h)^{-1}}. \end{aligned} \quad (45)$$

With the external labels suppressed, the recursion is

$$\begin{aligned} \mathfrak{F}_0(h; c) &= 1, \\ \mathfrak{F}_\ell(h; c) &= f_\ell(h) + \sum_{\substack{r \geq 2, \ s \geq 1 \\ r+s \in 2\mathbb{Z} \\ rs \leq 2\ell}} \frac{R_{r,s}^\ell(h)}{c - c_{r,s}(h)} \mathfrak{F}_{\ell-rs/2} \left(h + \frac{rs}{2}; c_{r,s}(h) \right). \end{aligned} \quad (46)$$

The endpoint $rs = 2\ell$ is essential: it supplies the first $(3, 1)$ pole at $\ell = 3/2$ and the first $(2, 2)$ pole at $\ell = 2$. BRY's displayed equation prints the strict inequality $rs < 2\ell$, whereas its ancillary notebook sums through $rs = 2\ell$. Equation (46) and the code follow the notebook.

At a pole set $b = b_{r,s}(h)$, $Q_{r,s} = b + b^{-1}$, and

$$a_i^{(r,s)} = \sqrt{\frac{Q_{r,s}^2}{4} - 2h_i}. \quad (47)$$

For $\delta \in \{0, 1\}$, define the fusion polynomial

$$P_{r,s}^{(\delta)}(h_a, h_b) = \prod_{\substack{p=1-r, 1-r+2, \dots, r-1 \\ q=1-s, 1-s+2, \dots, s-1 \\ (p+q)-(r+s) \equiv 2(1-\delta) \pmod{4}}} \frac{2a_a^{(r,s)} - 2a_b^{(r,s)} - pb - q/b}{2\sqrt{2}} \times \frac{2a_a^{(r,s)} + 2a_b^{(r,s)} + pb + q/b}{2\sqrt{2}}. \quad (48)$$

Let $\delta_i^{(\ell)} = \delta_i$ for integer ℓ and $\delta_i^{(\ell)} = 1 - \delta_i$ for half-integer ℓ . The residue is

$$R_{r,s}^\ell(h) = (-1)^{rs\delta_3} \left(-\frac{\partial c_{r,s}(h)}{\partial h} \right) A_{r,s}(h) P_{r,s}^{(\delta_2^{(\ell)})}(h_1, h_2) P_{r,s}^{(\delta_3^{(\ell)})}(h_4, h_3). \quad (49)$$

The factor $(-1)^{rs\delta_3}$ is required; omitting it is the sign typo in the older recursion formula noted and corrected by BRY.

For an implementation-complete seed ledger, set $A = h + h_3 - h_4$, $B = h + h_2 - h_1$. For $m \in \mathbb{Z}_{>0}$ and $n \in \mathbb{Z}_{\geq 0}$,

(δ_3, δ_2)	f_m	$f_{n+1/2}$
$(0, 0)$	$\frac{(A)_m (B)_m}{m! (2h)_m}$	$\frac{(A + \frac{1}{2})_n (B + \frac{1}{2})_n}{n! (2h)_{n+1}}$
$(0, 1)$	$\frac{(A)_m (B + \frac{1}{2})_m}{m! (2h)_m}$	$\frac{(A + \frac{1}{2})_n (B)_{n+1}}{n! (2h)_{n+1}}$
$(1, 0)$	$\frac{(A + \frac{1}{2})_m (B)_m}{m! (2h)_m}$	$-\frac{(A)_{n+1} (B + \frac{1}{2})_n}{n! (2h)_{n+1}}$
$(1, 1)$	$\frac{(A + \frac{1}{2})_m (B + \frac{1}{2})_m}{m! (2h)_m}$	$-\frac{(A)_{n+1} (B)_{n+1}}{n! (2h)_{n+1}}$

(50)

where $(x)_n$ is the rising Pochhammer symbol.

3.3 Elliptic representation and code interface

The local z -series is converted to the elliptic nome

$$q(z) = \exp \left[-\pi \frac{K(1-z)}{K(z)} \right], \quad K(z) = {}_2F_1 \left(\frac{1}{2}, \frac{1}{2}; 1; z \right), \quad z = \frac{\theta_2(q)^4}{\theta_3(q)^4}. \quad (51)$$

Both parities are written as

$$\mathcal{F}^{e/o}(z) = (16q)^{h-Q^2/8} z^{Q^2/8-h_1-h_2} (1-z)^{Q^2/8-h_2-h_3} \times \theta_3(q)^{\frac{3}{2}Q^2-4(h_1+h_2+h_3+h_4)} H^{e/o}(q). \quad (52)$$

Here H^e has integer powers of q , whereas H^o has powers $q^{n+1/2}$. The elliptic series converges rapidly throughout the direct channel and is the representation used for numerical four-punctured-sphere work.

The root-level Python module `superconformal_blocks.py` implements Eqs. (42)–(52). Its main class stores a level ℓ by the integer 2ℓ , exposes individual recursion coefficients, and evaluates either the local series or the elliptic series:

```

block = NSSphereFourPointBlock.from_liouville_momenta(
    p1=P1, p2=P2, p3=P3, p4=P4,
    internal_momentum=P, b=1, star2=True, star3=True)
Fe = block.elliptic_block(z, order=8, parity="even")
Fo = block.elliptic_block(z, order=8, parity="odd")

```

All square roots, fractional powers, and elliptic functions use principal branches. Crucially, the $a_i^{(r,s)}$ are recomputed with $Q_{r,s}$ at every recursion pole rather than with the physical background charge. A near-zero $c - c_{r,s}$ raises an explicit pole error instead of silently returning an unstable value. The order parameter counts retained coefficients: q^0 through q^{N-1} in the even block and $q^{1/2}$ through $q^{N-1/2}$ in the odd block.

As an external regression anchor, take BRY's ancillary-notebook values

$$c = \frac{27}{2} + 10^{-5}, \quad (P_4, P_3, P_2, P_1) = \left(\frac{3}{5}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}\right), \quad h = \frac{11}{10}, \quad (\delta_3, \delta_2) = (1, 1). \quad (53)$$

The implementation reproduces

$$\begin{aligned}
H^e(q) &= 1 + 1.0998737373737373q + 15.824538587885476q^2 + 21.09367085588721q^3 + \dots, \\
H^o(q) &= -1.7823926767676768q^{1/2} - 6.738809221289842q^{3/2} - 13.848138977969848q^{5/2} + \dots.
\end{aligned} \quad (54)$$

At $z = 1/3 + 3i/5$, truncation through eight coefficients gives

$$\begin{aligned}
\mathcal{F}^e &= 1.3109708952736965 + 0.18058411587390863i, \\
\mathcal{F}^o &= -0.3051714889777302 - 0.4379494400515853i.
\end{aligned} \quad (55)$$

The automated tests also compare the z - and q -representations for all eight choices and verify explicitly that both endpoint poles are present. No claim about a higher-genus analogue is made here; that literature and sewing audit is deferred to a later stage.

3.4 Nonchiral correlators and the BRY four-tachyon density

We now contract the chiral blocks with the $b = 1$ pants coefficients defined in Eqs. (26) and (27), using the explicit formulas (34) and (35). Place the operators 4, 3, 2, 1 at $\infty, 1, z, 0$, respectively, and abbreviate

$$C_{12P} = C(P_1, P_2, P), \quad C_{34P} = C(P_3, P_4, P), \quad \tilde{C}_{12P} = \tilde{C}(P_1, P_2, P), \quad \tilde{C}_{34P} = \tilde{C}(P_3, P_4, P). \quad (56)$$

Below, $\mathcal{F}^{e/o}$ has external labels (h_4, h_3, h_2, h_1) , whereas $\mathcal{F}_{**}^{e/o}$ has $(h_4, *h_3, *h_2, h_1)$. BRY's three nonchiral functions are

$$G(4321|z) = \int_0^\infty \frac{dP}{\pi} [C_{12P}C_{34P}\mathcal{F}^e(z)\mathcal{F}^e(\bar{z}) + \tilde{C}_{12P}\tilde{C}_{34P}\mathcal{F}^o(z)\mathcal{F}^o(\bar{z})], \quad (57)$$

$$H(4321|z) = - \int_0^\infty \frac{dP}{\pi} [\tilde{C}_{12P}\tilde{C}_{34P}\mathcal{F}_{**}^e(z)\mathcal{F}_{**}^e(\bar{z}) + C_{12P}C_{34P}\mathcal{F}_{**}^o(z)\mathcal{F}_{**}^o(\bar{z})]. \quad (58)$$

Here $G = \langle V_{P_4}(\infty)V_{P_3}(1)V_{P_2}(z, \bar{z})V_{P_1}(0) \rangle$ and $H = \langle V_{P_4}(\infty)W_{P_3}(1)W_{P_2}(z, \bar{z})V_{P_1}(0) \rangle$. The third function includes the two timelike-free-fermion contractions in the picture-raised tachyons:

$$\begin{aligned}
J(4321|z) &= - \int_0^\infty \frac{dP}{\pi} \left\{ C_{12P}C_{34P} \left[\frac{\mathcal{F}_{**}^o(z)\mathcal{F}^e(\bar{z})}{1-\bar{z}} + \frac{\mathcal{F}^e(z)\mathcal{F}_{**}^o(\bar{z})}{1-z} \right] \right. \\
&\quad \left. + \tilde{C}_{12P}\tilde{C}_{34P} \left[\frac{\mathcal{F}_{**}^e(z)\mathcal{F}^o(\bar{z})}{1-\bar{z}} + \frac{\mathcal{F}^o(z)\mathcal{F}_{**}^e(\bar{z})}{1-z} \right] \right\}.
\end{aligned} \quad (59)$$

In the scalar convention used in Eqs. (58) and (59), the doubly-starred odd block is minus the signed ancillary-notebook series returned by the low-level block class. This phase cancels between the two chiral halves of H but is indispensable in the mixed products of J . For nonnegative real momenta, H and J are real and negative, as required by the anti-Hermiticity of the descendant operators.

For the all-tachyon term in BRY's $1 \rightarrow 3$ setup, assign

$$(P_4, P_3, P_2, P_1) = (\omega, \omega_3, \omega_2, \omega_1). \quad (60)$$

After the internal- P integral but before the z integral, the reduced moduli density is

$$\mathcal{I}_T(z) = |z|^{-2\omega_1\omega_2} |1-z|^{-2\omega_2\omega_3} \left[\frac{\omega_2^2\omega_3^2}{|1-z|^2} G(4321|z) - H(4321|z) - \omega_2\omega_3 J(4321|z) \right]. \quad (61)$$

Thus the corresponding unregularized contribution carries the overall factor

$$\frac{i}{2} \delta \left(\omega - \sum_{i=1}^3 \omega_i \right) g_s^4 C_{S^2} \int_{\mathbb{C}} d^2 z \mathcal{I}_T(z). \quad (62)$$

This is the BRY expression after multiplying the explicitly calculated four-tachyon term by the eight nonzero even-axion contributions. Equality of those eight terms follows from the assumed perturbative decoupling of left- and right-side modes; it was not separately derived from eight worldsheet correlators.

The code interface is

```
corr = BRYNSFourPointCorrelator(
    p1=1/2, p2=1/3, p3=1/4, p4=3/5, block_order=8)
values = corr.evaluate(z=0.1, p_max=5, quadrature_order=20)
```

The module `sphere_four_point.py` performs the P integral by Gauss–Legendre quadrature, while `super_liouville_structure_constants.py` supplies the normalized C and $\tilde{C}$. The recursion is evaluated at $c = 27/2 + 10^{-5}$, as in BRY's numerical benchmark, and with arbitrary precision internally. The latter is necessary because separately large near-threshold recursion terms cancel at higher orders. The finite $P_{\max}$ is only a numerical truncation: convergence must be checked by increasing $P_{\max}$, the quadrature order, and the elliptic-block order.

At BRY's crossing-test momenta $(P_1, P_2, P_3, P_4) = (1/2, 1/3, 1/4, 3/5)$, using the parameters in the code snippet, the regression values are

$$G(0.1) = 0.6251752343, \quad H(0.1) = -0.3082787765, \quad J(0.1) = -0.4125437927. \quad (63)$$

With the same numbers interpreted as $(\omega_1, \omega_2, \omega_3, \omega)$, $\mathcal{I}_T(0.1) = 0.7630630252$. The automated tests check these anchors, the H crossing relation, structure-constant identities, and the near-threshold high-precision cancellation.

3.5 First regulated $1_R \rightarrow 3_R$ benchmark

The first end-to-end test follows BRY Eq. (4.15):

$$\omega = \frac{1}{3} + ia, \quad \omega_1 = \omega_2 = \omega_3 = \frac{\omega}{3}, \quad 0.5 \leq a \leq 0.7. \quad (64)$$

No pole of the super-Liouville structure constants crosses the undeformed contour $P \in \mathbb{R}_{\geq 0}$ along this continuation. The s - and u -channel counterterms vanish, as does the t -channel term proportional to $\tilde{C}\tilde{C}$. Only the leading CC subtraction at the picture-changing collision $z = 1$ remains. At fixed internal momentum its three picture-raised coefficients form the OPE square

$$\begin{aligned} B_t(P) &= \omega_2^2 \omega_3^2 + c_0(P)^2 + 2\omega_2 \omega_3 c_0(P) \\ &= \frac{1}{4} \left[1 + (\omega_2 + \omega_3)^2 - P^2 \right]^2, \quad c_0(P) = h_2 + h_3 - h_P. \end{aligned} \quad (65)$$

Consequently the nonzero fixed- P counterterm density, including dP/π , is

$$r_t(P, z) = \frac{1}{\pi} C(\omega_1, \omega, P) C(\omega_2, \omega_3, P) B_t(P) |1 - z|^{-3+P^2-(\omega_2+\omega_3)^2}, \quad 0 \leq P < P_*, \quad (66)$$

where

$$P_* = \sqrt{1 + \text{Re}[(\omega_2 + \omega_3)^2]}. \quad (67)$$

For $a = 0.6$, $P_* = 0.9430708966$. The subtraction is global, as required by analytic continuation of the power divergence; it is not a literal removal of only the $1/\epsilon$ term in a small disk.

Following BRY, inversion crossing folds the full z plane to $D = \{|z| \leq 1\}$, which is divided into

$$D_\epsilon^{(0)} = \{|z| < \epsilon\}, \quad D_{\text{int}} = \{\epsilon \leq |z| \leq 1, |1 - z| \geq \epsilon\}, \quad D_\epsilon^{(1)} = \{|z| \leq 1, |1 - z| < \epsilon\}. \quad (68)$$

We use $\epsilon = 10^{-2}$. The direct low- z OPE is integrated in $D_\epsilon^{(0)}$, the elliptic-recursion blocks are used in D_{int} , and the crossed low- $(1 - z)$ blocks through next-to-leading order are used in $D_\epsilon^{(1)}$. The angular quadrature in D_{int} is explicitly split at $\arg z = \pm\epsilon$; a single rule on $[-\pi, \pi]$ misses this narrow boundary layer.

Let $\mathcal{M}$ denote the regularized expression inside braces in BRY Eq. (B.6), after the three z integrals and the final P integral. The worldsheet and matrix-model normalizations imply

$$\mathcal{A}_{1_R \rightarrow 3_R}^{\text{WS}}|_{\delta, \mu_F^{-2} \text{ stripped}} = \frac{8i}{\pi} \mathcal{M}, \quad \mathcal{M}_{\text{MQM}} = \pi \omega \omega_1 \omega_2 \omega_3 (1 + 2i\omega), \quad (69)$$

which is equivalent to $\mathcal{A}_{1_R \rightarrow 3_R}^{\text{MQM}} = 8i\omega \omega_1 \omega_2 \omega_3 (1 + 2i\omega)$. The executable implementation is `bry_one_to_three.py`. It keeps the internal- P contour and threshold explicit, evaluates all three moduli regions, and reports both sides of Eq. (69) as complex numbers. Fast regression tests separately check the OPE square, P_* , the direct low- z expansion, complex external momenta, and the final normalization conversion and regularized integration path. At $a = 0.6$, a representative q^8 run with $P_{\text{max}} = 4$, 24-point P quadrature on each side of P_* , and the moduli orders recorded by the command-line output gives

$$\begin{aligned} \mathcal{M}_{\text{WS}} &= 0.01750879 - 0.00320541i, & \mathcal{M}_{\text{MQM}} &= 0.01772691 - 0.00297257i, \\ \mathcal{A}_{\text{WS}} &= 0.00816252 + 0.04458577i, & \mathcal{A}_{\text{MQM}} &= 0.00756958 + 0.04514122i. \end{aligned} \quad (70)$$

The relative complex-amplitude discrepancy is 1.78%. This recovers the BRY curve at the first nontrivial level, but does not yet reproduce their quoted sub-0.3% numerical precision. At the same production quadrature, the cutoff comparison begins with

$$\begin{aligned} \mathcal{M}_{q^6} &= 0.01765123551 - 0.00248615555i, & \mathcal{M}_{q^8} &= 0.01750879244 - 0.00320541444i, \\ \mathcal{M}_{q^{10}} &= 0.01750781745 - 0.00322155122i. \end{aligned} \quad (71)$$

In units of $|\mathcal{M}_{\text{MQM}}|$, the $q^6 \rightarrow q^8$ displacement is 4.08%, while the $q^8 \rightarrow q^{10}$ displacement is approximately 0.09%; changing the moduli orders from 12 to 14 at q^8 gives 0.074%. Thus q^6 is not converged and production begins at approximately q^8 . The remaining limitation is the narrow- $z = 1$ and oscillatory- P product quadrature rather than the superconformal recursion.

4 Matrix quantum mechanics

4.1 Double-scaled Hamiltonian

The BRY double-scaled matrix quantum mechanics is

$$H = \frac{1}{2} \text{Tr}(P^2 - X^2), \quad h = \frac{1}{2}(p^2 - x^2). \quad (72)$$

Diagonalization and the Vandermonde determinant turn the eigenvalues into free fermions. All one-particle states on both sides with

$$E \leq -\mu_F, \quad \mu_F > 0, \quad (73)$$

are filled. No extra oscillator curvature, matrix Planck constant, or α' is to be restored in Eq. (72).

4.2 Two-sided basis and exact dictionary

BRY's physical left- and right-side scattering modes are

$$\mathcal{L}_\omega^\pm = T_\omega^\pm - A_\omega^\pm, \quad \mathcal{R}_\omega^\pm = T_\omega^\pm + A_\omega^\pm. \quad (74)$$

There is no $1/\sqrt{2}$ in this basis. Reflection exchanges $\mathcal{L}$ and $\mathcal{R}$ and realizes $(-1)^{F_L}$; hence T is even and A is odd. At zero flux, correlators with an odd number of axions vanish.

The two sides decouple to every order in the perturbative $1/\mu_F$ expansion. Eigenvalue tunnelling across the barrier mixes them through effects nonanalytic in g_s . Perturbative and tunnelling contributions must therefore be tracked separately.

For one incoming and k outgoing excitations on one side,

$$\mathcal{A}_{1 \rightarrow k} = \sum_{g \geq 0} \mu_F^{-(k-1+2g)} \mathcal{A}_{\text{pert}}^{(g)} + O(e^{-2\pi\mu_F}). \quad (75)$$

The exact worldsheet/matrix-model dictionary is

$$\boxed{\omega_{\text{MM}} = 2\omega, \quad g_s = \frac{4}{\pi\mu_F}, \quad C_{S^2} = \frac{\pi}{g_s^2}.} \quad (76)$$

The energy relation includes the corresponding rescaling of the energy-conservation delta function. In the phases of Eqs. (16)–(18), BRY compare directly to the matrix-model S-matrix; an additional leg-pole transform must not be inserted.

5 Duality dictionary

Equation (74), rather than an independently orthonormalized density basis, is used in every amplitude comparison.

6 Genus-one path integral in BRY variables

We use the term free energy for the connected string vacuum functional, $\mathcal{F} = \log Z_{\text{string}}$; it is not the thermodynamic quantity $-\beta^{-1} \log Z$. This sentence fixes the overall sign convention.

String description	Matrix-model description
Liouville coordinate	Eigenvalue time-of-flight coordinate
BRY Fermi parameter μ_F	Depth of the Fermi level below the quadratic maximum
$g_s = 4/(\pi\mu_F)$	Expansion about the filled two-sided Fermi sea
NS–NS scalar T	Even combination of left/right scattering modes
R–R axion A	Odd combination of left/right scattering modes
Worldsheet energy ω	Matrix energy $\omega_{\text{MM}} = 2\omega$
$(-1)^{F_L}$	Reflection $M \mapsto -M$
R–R background data	Fermi-level asymmetry and flux data
D0-brane excitation	A fermion excited from the sea toward the top
Nonperturbative correction	Eigenvalue tunnelling across the barrier

Table 1: Baseline 0B/matrix-model dictionary. Flux normalizations are deferred until the flux milestone.

Define the physical radius and its dimensionless BRY numerical value by

$$\rho \equiv \frac{R_{\text{phys}}}{\ell_B}, \quad x \equiv \frac{R_{\text{phys}}}{\sqrt{2\alpha'}} = \frac{\rho}{2}, \quad y \equiv \frac{R_{\text{phys}}}{\sqrt{\alpha'}} = \frac{\rho}{\sqrt{2}} = \sqrt{2}x. \quad (77)$$

The variable y is the argument used in the finite-radius formulas of Ref. [1]. We denote the connected amplitude per regulated BRY Liouville-coordinate length by $\hat{\mathcal{F}}_1 = \mathcal{F}_1/V_\phi^{\text{reg}}$.

6.1 Even spin structures and the modular orbit integral

We follow the worldsheet derivation of Ref. [1], translated to Eq. (77). For an even spin structure $(r, s) = (0, 1), (1, 0), (1, 1)$, let $D_{r,s}$ be the fermion determinant divided by the square root of the scalar determinant, and define

$$\Theta_\rho(\tau) = \sum_{m,n \in \mathbb{Z}} e^{-S_{m,n}}, \quad S_{m,n} = \frac{\pi\rho^2}{2\tau_2} |n - m\tau|^2. \quad (78)$$

The compact matter multiplet, super-Liouville theory, and combined ghost system contribute respectively

$$Z_{r,s}^X = \frac{y}{\sqrt{\tau_2}} |D_{r,s}|^2 \Theta_\rho, \quad Z_{r,s}^{\text{SL}} = \frac{V_\phi^{\text{reg}}}{2\pi\sqrt{2\tau_2}} |D_{r,s}|^2, \quad Z_{r,s}^{\text{gh}} = \frac{1}{2\tau_2} |D_{r,s}|^{-4}. \quad (79)$$

All nonzero-mode determinants cancel. Including the diagonal GSO factor $1/2$ for each of the three even structures gives

$$\hat{\mathcal{F}}_{1,\text{even}} = \frac{3\rho}{16\pi} \int_{\mathcal{F}} \frac{d^2\tau}{\tau_2^2} \Theta_\rho(\tau). \quad (80)$$

This modular integral can be evaluated without a divergent vacuum-energy sum. Separate the zero orbit and unfold every nonzero orbit to the strip:

$$\begin{aligned} I(\rho) &\equiv \int_{\mathcal{F}} \frac{d^2\tau}{\tau_2^2} \Theta_\rho(\tau) \\ &= \frac{\pi}{3} + 2 \sum_{k=1}^{\infty} \int_{-1/2}^{1/2} d\tau_1 \int_0^\infty \frac{d\tau_2}{\tau_2^2} \exp \left[-\frac{\pi\rho^2 k^2}{2\tau_2} \right] \\ &= \frac{\pi}{3} + \frac{4}{\pi\rho^2} \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi}{3} \left(1 + \frac{2}{\rho^2} \right). \end{aligned} \quad (81)$$

The series is absolutely convergent; the last equality uses the ordinary Basel sum $\sum_{k \geq 1} k^{-2} = \pi^2/6$, not analytic continuation at $s = -1$. Therefore

$$\boxed{\hat{\mathcal{F}}_{1,\text{even}} = \frac{\rho}{16} + \frac{1}{8\rho}}. \quad (82)$$

6.2 Odd spin structure and direct odd-supermodulus integration

The odd structure $(r, s) = (0, 0)$ has left- and right-moving γ zero modes from the conformal Killing spinors, β zero modes from the gravitino components that cannot be gauged away, and zero modes for both the matter fermion χ and Liouville fermion ψ . The Liouville-fermion zeros cancel the γ -zero-mode divergence. Integrating the β zeros, equivalently the odd moduli, inserts the supercurrents

$$G(z)\bar{G}(\bar{w}) = (\chi\partial X + \psi\partial\phi)(z)(\bar{\chi}\bar{\partial}X + \bar{\psi}\bar{\partial}\phi)(\bar{w}). \quad (83)$$

These insertions absorb the remaining $\chi, \bar{\chi}$ zero modes. The nontrivial matter factor is then

$$\left\langle \partial X(z)\bar{\partial}X(\bar{w}) \right\rangle_X, \quad (84)$$

including both its separated-point term $-\pi/\tau_2$ and its contact term. Because the compact-boson action is proportional to $\int \partial X \bar{\partial} X$, the integrated insertion is a radius derivative of the circle path integral. It fixes the odd contribution to be proportional to $y - y^{-1}$.

Equivalently, one can work globally on the odd supertorus

$$(z, \theta) \sim (z + 1, \theta) \sim (z + \tau + \lambda\theta, \theta + \lambda) \quad (85)$$

and integrate $d\lambda d\bar{\lambda}$ directly. Thus no arbitrary local PCO position is introduced. If a local PCO description is used instead, it must reproduce this direct integral and its contact term; there is no additional vertical correction to add to a globally defined supermoduli integral for this unpunctured genus-one vacuum amplitude.

The path integral fixes the radius dependence. As in Ref. [1], its overall magnitude is normalized by the large-radius spectrum. The result before choosing the 0A/0B sign is

$$\hat{\mathcal{F}}_{1,\text{odd}}^{(\varepsilon)} = \frac{\varepsilon}{24\sqrt{2}} \left(y - \frac{1}{y} \right) = \varepsilon \left(\frac{\rho}{48} - \frac{1}{24\rho} \right). \quad (86)$$

The sign is chosen so that the $1/R$ term counts one massless field in 0A and two in 0B. In this convention $\varepsilon_A = +1$ and $\varepsilon_B = -1$, hence

$$\boxed{\hat{\mathcal{F}}_{1,\text{odd}}^{0B} = -\frac{\rho}{48} + \frac{1}{24\rho}}. \quad (87)$$

6.3 The 0B and fluxless 0A spin sums

The even contribution in Eq. (82) is common to the two theories. Their relative GSO choice reverses the odd contribution, so

$$\hat{\mathcal{F}}_{1,\text{odd}}^{0A}(\rho_A) = \frac{\rho_A}{48} - \frac{1}{24\rho_A} = -\hat{\mathcal{F}}_{1,\text{odd}}^{0B}(\rho_A). \quad (88)$$

Adding even and odd structures gives directly

$$\boxed{\hat{\mathcal{F}}_1^{0B}(\rho_B) = \frac{\rho_B}{24} + \frac{1}{6\rho_B}} \quad (89)$$

$$\boxed{\hat{\mathcal{F}}_1^{0A}(\rho_A) = \frac{1}{12} \left(\rho_A + \frac{1}{\rho_A} \right)}. \quad (90)$$

The BRST spectrum explains these coefficients: 0B has two momentum towers and one winding tower, while 0A has one momentum tower and two winding towers. This is now a check of the modular and supermoduli derivation, not a zeta-regularized input.

The physical T-duality map is

$$R_A = \frac{\alpha'}{R_B}, \quad \rho_A = \frac{2}{\rho_B}, \quad \hat{\mathcal{F}}_1^{0A}(\rho_A) = \hat{\mathcal{F}}_1^{0B}\left(\frac{2}{\rho_A}\right). \quad (91)$$

This also maps the 0B stationary point $\rho_B = 2$ to the 0A stationary point $\rho_A = 1$, or $R_A = \sqrt{\alpha'/2}$.

6.4 Liouville wall and the regulator ledger

The interaction in Eq. (4) is invariant under

$$\phi \longrightarrow \phi + c, \quad \mu_L \longrightarrow e^{-bc} \mu_L. \quad (92)$$

Consequently the wall position contains $-b^{-1} \log \mu_L$. With a fixed coordinate cutoff in the weak-coupling region, define

$$V_\phi^{\text{reg}} = -\frac{1}{b} \log \frac{\mu_L}{\Lambda_L} + V_{\text{scheme}}. \quad (93)$$

The additive term depends on the location and shape of the cutoff. It is not compared. Combining Eqs. (89) and (93) at $b = 1$ gives

$$\boxed{\mathcal{F}_1^{0B}(\rho, \mu_L)|_{\text{univ}} = -\frac{\log(\mu_L/\Lambda_L)}{24} \left(\rho + \frac{4}{\rho} \right)}. \quad (94)$$

For fluxless type 0A, the same regulated volume gives

$$\boxed{\mathcal{F}_1^{0A}(\rho_A, \mu_L)|_{\text{univ}} = -\frac{1}{12} \left(\rho_A + \frac{1}{\rho_A} \right) \log \frac{\mu_L}{\Lambda_L}}. \quad (95)$$

Restoring dimensions, this is

$$\mathcal{F}_1^{0A}|_{\text{univ}} = -\frac{\log(\mu_L/\Lambda_L)}{12\sqrt{2}} \left(\frac{2R_A}{\sqrt{\alpha'}} + \frac{\sqrt{\alpha'}}{R_A} \right). \quad (96)$$

The regulator-independent datum is the logarithmic derivative

$$\boxed{\frac{\partial \mathcal{F}_1^{0B}}{\partial \log \mu_L} = -\frac{1}{24} \left(\rho + \frac{4}{\rho} \right)}. \quad (97)$$

If another source uses $\Phi = a\phi$ and a parameter λ defined by $\mu_L = \kappa\lambda^p$, then

$$V_\Phi^{\text{reg}}|_{\log \lambda} = -\frac{ap}{b} \log \lambda. \quad (98)$$

Thus κ and additive cutoff changes affect only a constant, whereas a field rescaling a or power map p changes the logarithmic coefficient.

6.5 Douglas convention and the matrix variable

The calculation above follows Sec. 4.2 and App. A of Ref. [1], with $y = R_{\text{phys}}/\sqrt{\alpha'}$ translated to the BRY radius $\rho = \sqrt{2}y$. That reference writes the Liouville kinetic term as $(8\pi)^{-1}(d\phi)^2$, which is the same coordinate normalization as Eq. (4), and then sets $V_L = -\log|\mu_D|$, with an additive cutoff constant suppressed. At $b = 1$ this is Eq. (93) after a constant rescaling of the interaction parameter. Their Eqs. (4.5), (4.7), and (4.8) become respectively Eqs. (82), (86), and the two totals above.

To compare the logarithm, introduce rather than assume

$$\mu_L = \kappa \mu_F^p. \quad (99)$$

Equation (94) then predicts

$$\mathcal{F}_1^{0B}|_{\log \mu_F} = -\frac{p}{24} \left(\rho + \frac{4}{\rho} \right) \log \mu_F. \quad (100)$$

The conventional string/matrix-model identification corresponds to $p = 1$ and an unobservable constant κ . Agreement therefore tests $p = 1$; it is not an input to the worldsheet moduli integral.

For the smoke-test choice $\mu_L/\Lambda_L = 10$, the stationary numerical radius is $\rho = 2$, equivalently $R_{\text{phys}} = \sqrt{2\alpha'}$, and

$$\left. \frac{\partial \mathcal{F}_1^{0B}}{\partial \log \mu_L} \right|_{\rho=2} = -\frac{1}{6}, \quad \mathcal{F}_1^{0B}(\rho=2)|_{\text{univ}} = -\frac{\log 10}{6}. \quad (101)$$

7 Benchmark ladder

The calculations should be developed in the following order.

A. Analytic torus check

First reproduce the determinant cancellation in Eq. (79), unfold the convergent modular integral Eq. (81), and derive the odd-spin term by integrating the odd supermodulus with its contact term. Then apply the 0A/0B sign and the Liouville wall displacement. The spectrum count and matrix variable μ_F are subsequent checks.

B. Inverted-oscillator scattering data

Construct even and odd one-particle scattering states, compute their reflection phases and regulated densities of states, and reconstruct the finite-radius genus-one term from the two parity sectors. Cutoff variation and increasing numerical precision should leave the logarithmic coefficient stable.

C. Collective fields and sphere amplitudes

Build the left and right time-of-flight fields, form the BRY combinations $\mathcal{L} = T - A$ and $\mathcal{R} = T + A$, verify the zero-flux parity selection rule, and reproduce the BRY-normalized $1 \rightarrow 2$ and $1 \rightarrow 3$ amplitudes with $\omega_{\text{MM}} = 2\omega$. Mixed-side connected amplitudes must vanish perturbatively.

D. Flux backgrounds and 0A/0B T-duality

Introduce asymmetric Fermi data using explicitly named flux variables. Keep Lorentzian continuous-flux conventions distinct from Euclidean quantized fluxes, and verify the exact finite-radius 0A/0B map [8, 9].

E. Higher genus and StringMC

Only after the lower-genus dictionary is fixed should the project choose a super-Riemann-surface measure, specify the treatment of odd moduli, and adapt the existing StringMC sampling infrastructure. The bosonic moduli measure must not be imported into this target implicitly. Degeneration, factorization, GSO phases, and superghost normalization must be audited before a full higher-genus integral is attempted.

8 Worksheet literature for the genus-one and genus-two program

The higher-genus target requires more than a bosonic moduli integral with a spin-structure prefactor. It needs the BRST-complete state space, the bc - $\beta\gamma$ measure, the integration over odd moduli, and the super-Liouville conformal data in every allowed NS and Ramond sewing channel. A detailed source-to-formula map is maintained in `literature_genus1_genus2.md`; the main conclusions are summarized here.

Ref. [3] computes tree-level sphere $1 \rightarrow 2$ and $1 \rightarrow 3$ amplitudes. It does not compute the torus vacuum amplitude or the genus-two free energy, and explicitly leaves higher-genus blocks and loop amplitudes for future work. We therefore use it as the master convention and sphere-data source, not as a substitute for the higher-genus measure.

8.1 Genus-one measure source

The generic propagating spectrum consists of the NS–NS tachyon and R–R axion, supplemented by discrete BRST states at special momenta. The physical 0B spectrum and its circle momentum/winding projection are given in Ref. [1]; the systematic NS and R BRST cohomology is given in Ref. [10].

Section 5 follows the explicit genus-one measure of Ref. [1], translated to the BRY ledger. For the three even torus spin structures, the matter, super-Liouville, and combined superghost factors are

$$\begin{aligned} Z_X &= \frac{y}{\sqrt{\tau_2}} |D_{r,s}|^2 \sum_{m,n} e^{-S_{m,n}}, \\ Z_{\text{SL}} &= \frac{V_D}{2\pi\sqrt{2\tau_2}} |D_{r,s}|^2, & Z_{\text{sgh}} &= \frac{1}{2\tau_2} |D_{r,s}|^{-4}. \end{aligned} \quad (102)$$

All nonzero-mode determinants cancel. An even supertorus has moduli dimension $1|0$, so there is no PCO insertion for the unpunctured vacuum amplitude. The quantity V_D is their regulated Liouville volume; it is translated using Eq. (98), rather than identified by its symbol.

The odd spin structure must not be omitted. It has one odd supermodulus, one β and one γ zero mode in each chiral sector, and matter and Liouville fermion zero modes. Appendix A of Ref. [1] integrates the odd modulus directly. The β zero modes insert $G\bar{G}$, the Liouville-fermion zero modes cancel the γ -zero-mode divergence, and the remaining insertion is proportional to $\langle \partial X \bar{\partial} X \rangle$, including its contact term. This gives the odd-spin contribution required for Eq. (94).

For this vacuum calculation, direct integration over the global odd supertorus modulus is the primary prescription. Vertical integration is needed only if the same integral is represented by local PCO sections, or if punctures or off-shell states destroy the global description. It must then reproduce, rather than be added to, the direct-supermoduli result.

8.2 Genus two: conformal data and measure

For an unpunctured surface of genus $g \geq 2$,

$$\dim \mathfrak{M}_g = (3g - 3 | 2g - 2). \quad (103)$$

Thus the genus-two chiral measure has dimension $3|2$: a PCO representation uses two holomorphic and two antiholomorphic PCOs. There are ten even and six odd genus-two spin structures [11]. The all-genus type-0 GSO projection is a diagonal spin-structure sum. Type 0B assigns no Arf sign, whereas 0A weights a spin structure by $(-1)^{\text{Arf}}$ [12].

The complete three-point coefficient is a family of coefficients. The NS sector has two independent structures,

$$C(P_1, P_2, P_3), \quad \tilde{C}(P_1, P_2, P_3), \quad (104)$$

and an R–R–NS pants vertex requires two further structures, which may be taken as C_{even} and C_{odd} . They were bootstrapped in Refs. [13, 14]. For numerical work at $b = 1$, the preferred normalization is the explicit $\Upsilon_{\text{NS/R}}$ representation of Ref. [3], whose formulas and reflection rules have already been fixed explicitly in Eqs. (30)–(37). They are part of the initial worldsheet convention, not data introduced only at genus two. In particular, PCO supercurrents couple to top components and descendants, so the bottom coefficient C alone is insufficient.

The structure constants must be combined with super-Virasoro sewing blocks. NS elliptic recursions, including top-component blocks, are developed in Refs. [15, 6, 7]; Ramond blocks are treated in Ref. [16]. These are benchmarks for a plumbing-fixture sewing engine, not a ready-made genus-two vacuum block.

For even spin structures, the natural starting point is the slice-independent D’Hoker–Phong measure based on the super period matrix [17, 18]. Their formula for a general $N = 1$, $c = 15$ matter SCFT contains both the matter-supercurrent correlator and the finite-dimensional b - and β -ghost determinants. The latter are essential: the apparent collision pole of the naive product of two PCOs is cancelled by an opposite singularity in those determinants. For the present problem, the flat-space matter partition function must be replaced by the noncompact time $\times$ super-Liouville theory, with its Liouville zero-mode regulator and degeneration boundary terms retained.

The standard explicit D’Hoker–Phong result is an even-spin construction. The six odd spin structures are therefore a separate work package. Their fermion and superghost zero modes, contact terms, and separating and nonseparating degenerations must be derived directly before translating the answer into a PCO gauge.

The BRY coupling relation does fix the conversion of genus powers,

$$\frac{1}{\mu_F^2} = \frac{\pi^2}{16} g_s^2. \quad (105)$$

Consequently a matrix-model term A_2/μ_F^2 maps to $(\pi^2 A_2/16)g_s^2$. This power map does not determine the absolute unpunctured genus-two path-integral normalization. That constant must be fixed by degeneration and factorization to the BRY-normalized sphere data and the torus anchor.

8.3 Vertical integration at genus two

If PCOs are used globally, spurious poles obstruct a single smooth PCO section. Sen and Witten prescribe safe local sections and vertical interpolations between them [19, 20]. With two PCOs, the construction contains bulk cell integrals, codimension-one vertical segments on shared faces, and codimension-two vertical cells at corners; the hierarchy then terminates. The choices must also respect plumbing degenerations. This is an alternative representation of the odd-moduli integral, not an additional physical correction to a correct super-period-matrix calculation.

8.4 Result of the literature audit

The primary literature supplies the complete sphere super-Liouville three-point data and a powerful even-spin genus-two gauge-fixing framework. It does not appear to supply the finished diagonal type-0B genus-two vacuum integrand for time $\times$ super-Liouville including all odd spin structures. Constructing and factorization-testing that assembly is the main worldsheet research problem for the genus-two target.

9 Deferred choices and safeguards

The baseline intentionally leaves the following items open:

- continuation to $\mu_F < 0$ through the top of the potential;
- unequal filling and the two independent 0B flux variables;
- Lorentzian versus Euclidean flux normalization;
- the complete 0A/0B T-duality dictionary;
- the nonperturbative tunnelling contour prescription;
- the higher-genus supermoduli integration cycle and measure.

An imbalance between the Fermi seas is not a harmless numerical regulator: it changes the R–R background. Likewise, Eq. (76) fixes the sphere coupling convention but does not by itself normalize the unpunctured genus-two integration cycle. Both facts should remain explicit in code and data products.

10 Immediate next step

The next bounded calculation is to complete stage A at the integrand level: derive Eq. (87) by direct integration of the odd supertorus modulus, including its contact term, without importing its value from Douglas et al. The matrix-model phase-shift calculation then tests Eq. (99) and in particular the exponent p .

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